Week 5 | Solve With Instructors

Q1) Suppose you are given three training instance $(\mathbf{x_1}, y_1)$, $(\mathbf{x}_2, y_2)$ and $(\mathbf{x}_3, y_3)$ as follows:

$$(\mathbf{x}_1, y_1) = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, 1\right),\ (\mathbf{x}_2, y_2) = \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}, 1\right), (\mathbf{x}_3, y_3) = \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, 0\right),$$

where each x_i is a data point and y_i is its corresponding label. Your goal is to predict the output $\widehat{\mathbf{y}}$ based on the features where $\widehat{\mathbf{y}} = \mathbf{X}^T\widehat{\mathbf{w}}$ where $\widehat{\mathbf{w}} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$. Find the $\widehat{\mathbf{w}}$ that minimizes $||\widehat{\mathbf{y}} - \mathbf{y}||^2$ where $\mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$.

Hint: Use the below formula to calculate the inverse of a 2×2 matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Solution:

Given matrix $\mathbf{X} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ and label vector $\mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$,

$$\mathbf{X}\mathbf{X}^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\left(\mathbf{X}\mathbf{X}^T\right)^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Using normal equations

$$\widehat{\mathbf{w}} = \left(\mathbf{X}\mathbf{X}^T\right)^{-1} \mathbf{X}\mathbf{y}$$

$$= \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Q2) For the dataset given below

$$\mathbf{X}_{d\times n} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{bmatrix}, \ \mathbf{y} = \begin{bmatrix} 2 \\ 3 \\ 1 \\ 3 \end{bmatrix}$$

Initialize $\mathbf{w}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ be the initial vector with learning rate $\eta = 0.01$. Let $\mathbf{w}_{\text{gd}}$ be the updated weight after one iteration of gradient descent and $\mathbf{w}_{\text{sgd}}$ be the updated weight after one full iteration of Stochastic gradient descent by taking first two and then next two data point for the sample. Compute $\|\mathbf{w}_{\text{gd}} - \mathbf{w}_{\text{sgd}}\|$. For all the calculations, consider $\frac{1}{2}\left\|\mathbf{X}^T\mathbf{w} - \mathbf{y}\right\|^2$ as your loss function.

Solution:

Gradient Descent

$$\mathbf{w}_{\text{gd}} = \mathbf{w}_0 - \eta\left(\mathbf{X}\mathbf{X}^T\mathbf{w}_0 - \mathbf{X}\mathbf{y}\right)$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.01\left(\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 3 & 4 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{bmatrix}\begin{bmatrix} 2 \\ 3 \\ 1 \\ 3 \end{bmatrix}\right)$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.01\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 23 \\ 20 \end{bmatrix}\right)$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.23 \\ 0.20 \end{bmatrix} = \begin{bmatrix} 0.23 \\ 0.20 \end{bmatrix}$$

Stochastic Gradient Descent

➔ For the first two data points

$$\mathbf{w}_{\text{sgd}}^{(1)} = \mathbf{w}_0 - \eta\left(\mathbf{X}_{\text{batch1}}\mathbf{X}_{\text{batch1}}^T\mathbf{w}_0 - \mathbf{X}_{\text{batch1}}\mathbf{y}\right)$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.01\left(\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right)$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 0.01\begin{bmatrix} 8 \\ 7 \end{bmatrix}$$

$$= \begin{bmatrix} 0.08 \\ 0.07 \end{bmatrix}$$

➔ For the next two data points

$$\mathbf{w}_{\text{sgd}}^{(2)} = \mathbf{w}_{\text{sgd}}^{(1)} - \eta\left(\mathbf{X}_{\text{batch2}}\mathbf{X}_{\text{batch2}}{}^{T}\mathbf{w}_{\text{sgd}}^{(1)} - \mathbf{X}_{\text{batch2}}\mathbf{y}\right)$$

$$= \begin{bmatrix} 0.08 \\ 0.07 \end{bmatrix} - 0.01\left(\begin{bmatrix} 3 & 4 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} 3 & 4 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} 0.08 \\ 0.07 \end{bmatrix} - \begin{bmatrix} 3 & 4 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right)$$

$$= \begin{bmatrix} 0.08 \\ 0.07 \end{bmatrix} - 0.01\left(\begin{bmatrix} 3.68 \\ 3.67 \end{bmatrix} - \begin{bmatrix} 15 \\ 13 \end{bmatrix}\right)$$

$$= \begin{bmatrix} 0.08 \\ 0.07 \end{bmatrix} + 0.01\begin{bmatrix} 11.32 \\ 9.33 \end{bmatrix}$$

$$= \begin{bmatrix} 0.1932 \\ 0.1632 \end{bmatrix}$$

Final $\mathbf{w}_{\text{sgd}}$ is given by $\frac{1}{2}\left(\mathbf{w}_{\text{sgd}}^{(1)} + \mathbf{w}_{\text{sgd}}^{(2)}\right)$.

$$\begin{aligned}
\mathbf{w}_{\text{sgd}} &= \frac{1}{2}\left(\mathbf{w}_{\text{sgd}}^{(1)} + \mathbf{w}_{\text{sgd}}^{(2)}\right) \\
&= \frac{1}{2}\left(\begin{bmatrix} 0.08 \\ 0.07 \end{bmatrix} + \begin{bmatrix} 0.1932 \\ 0.1632 \end{bmatrix}\right) \\
&= \begin{bmatrix} 0.1366 \\ 0.1166 \end{bmatrix}
\end{aligned}$$

Computing $||\mathbf{w}_{\text{gd}} - \mathbf{w}_{\text{sgd}}||$

$$\left\| \begin{bmatrix} 0.23 \\ 0.20 \end{bmatrix} - \begin{bmatrix} 0.1366 \\ 0.1166 \end{bmatrix} \right\|$$

$$\approx 0.073$$

Q3) Consider the following data-points for a regression problem:

$$\{(c_1 \cdot \mathbf{u}, y_1), \ \cdots, (c_n \cdot \mathbf{u}, y_n)\}$$

Here, c_i is some real number and $\mathbf{u} \in \mathbb{R}^d$ and $\mathbf{u} \neq 0$. Fit a linear regression model for this dataset and find the predicted value for the test-point $x_{\text{test}} = 5 \cdot \mathbf{u}$. You can assume that $c_i \neq 0$ for some i. The following values are given to you:

$$\sum_{i=1}^{n} c_i \cdot y_i = 20, \ \ \sum_{i=1}^{n} c_i^2 = 100$$

Solution: The loss is

$$L(\mathbf{w}) = \sum_{i=1}^{n} \left(\mathbf{w}^T \mathbf{x}_i - y_i\right)^2$$

The gradient is

$$\nabla L(\mathbf{w}) = 2 \cdot \sum_{i=1}^{n} \left(\mathbf{w}^T \mathbf{x}_i - y_i\right) \mathbf{x}_i$$

Replace $\mathbf{x}_i$ with $c_i \cdot \mathbf{u}$

$$\nabla L(\mathbf{w}) = 2 \cdot \sum_{i=1}^{n} \left(\mathbf{w}^T c_i \cdot \mathbf{u} - y_i\right) c_i \cdot \mathbf{u}$$

$$= 2 \cdot \sum_{i=1}^{n} \left[\left(\mathbf{w}^T \mathbf{u}\right) c_i^2 - c_i y_i \right] \mathbf{u}$$

On setting $\nabla L(\mathbf{w})$ to zero, we get

$$\mathbf{w}^T \mathbf{u} = \frac{\sum_{i=1}^{n} c_i y_i}{\sum_{i=1}^{n} c_i^2}$$

$$= 0.2$$

Now, the prediction for a point $5 \cdot \mathbf{u}$ is $\mathbf{w}^T(5 \cdot \mathbf{u})$ which turn out to be 1.

Q4) Kernel regression with a polynomial kernel of degree 3 is applied on a dataset $\{X, y\}$. Let the weight vector be given by

$$w = \phi(X)\begin{bmatrix} 1.3 & 0.6 & -0.2 & -0.7 \end{bmatrix}^T$$

Here $\phi(X)$ is the transformed data matrix whose i^{th} column is $\phi(x_i)$. What will be the prediction for the data point $\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}^T$?

Solution:

The prediction of test data point in kernel regression is

$$\sum_{i=1}^{n} \alpha_i \cdot k(\mathbf{x}_{\text{test}}, \mathbf{x}_i)$$

Given kernel function is

$$k(\mathbf{x}, \mathbf{y}) = \left(1 + \mathbf{x}^T \mathbf{y}\right)^3$$

Now, as our test point is $\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}^T$, the dot product term in kernel function will be zero with each data point. So, for each data
point $\mathbf{x}_i$ the value of kernel function would be just 1. Hence, prediction will be

$$\sum_{i=1}^{n} \alpha_i \cdot k(\mathbf{x}_{\text{test}}, \mathbf{x}_i) = 1.3(1) + 0.6(1) - 0.2(1) - 0.7(1)$$

$$= 1$$

$Q5$) Consider the following data points.

$$\{(1,2),(3,1),(-1,4)\}$$

The projection of the label vector onto the span of the feature vectors of the data points is $(2,11,-8)$. Predict the label of for $(7,3)$ using linear regression.

ANSWER : 25

The projection of the label vectors onto the span of the feature vectors of the data points is also the prediction for the labels of the data points.

$$X^T w = y_{pred}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 1 \\ -1 & 4 \end{bmatrix} \times \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 11 \\ -8 \end{bmatrix}$$

Solve the linear equations to get $w_1 = 4$ and $w_2 = -1$.

The prediction of label for $(7, 3)$ is given as :

$$(7, 3) \cdot (4, -1) = 25$$

$Q6$) We have the following data points

$$D = \{(x_1, y_1), \ldots, (x_n, y_n)\}$$

$$x_1, \ldots, x_n \in \mathbb{R}^d$$
$$y_1, \ldots, y_n \in \mathbb{R}$$

We get a w after linear regression such that :

$$\sum_{i=1}^{n} (y_i - f(x_i))^2 = 0$$

where $f(x_i) = w^T x_i$

Which of the following options are true

A. w is a result of overfitting
B. Every datapoint in D is on some hyperplane in $\mathbb{R}^d$
C. Every datapoint in D is on some hyperplane in $\mathbb{R}^{d+1}$
D. Let $Y = (y_1, \ldots, y_n)^T$ and let Y_p be the projection of Y onto the span of the feature vectors of $x_1, \ldots, x_n$. Then $||Y - Y_P|| > 0$

ANSWER : C

It is given that the squared error for linear regression is 0. This would mean that there is actually a linear relationship between the features and the labels. Here, $f(x_i)$ is not memorizing the labels of every data point. Therefore, we cannot say that w is a result of overfitting.

Since the error is zero, every point in D will be on a hyperplane in $\mathbb{R}^{d+1}$.

The projection of labels onto the feature space is the prediction for the labels. Since the error is 0, Y_p will be the same as Y, therefore, $||Y - Y_P|| = 0$

$Q7)$ $f(x) = x^2 - 2x + 1$. We are trying to find an x at which the function's value is minimum. We will use gradient descent by initializing $x_0 = 0$. After how many iterations does the algorithm stop? Use $\eta = 0.5$

ANSWER : 2

$f'(x) = 2x - 2$

<u>ITERATION 1</u>

$$\begin{aligned} x_1 &= x_0 - \eta \bigtriangledown f(x_0) \\ &= 0 - 0.5 \times f'(0) \\ &= 0 - 0.5 \times (-2) \\ &= 1 \end{aligned}$$

<u>ITERATION 2</u>

$$\begin{aligned} x_2 &= x_1 - \eta \bigtriangledown f(x_1) \\ &= 1 - 0.5 \times f'(1) \\ &= 1 - 0.5 \times 0 \\ &= 1 \end{aligned}$$

$x_2 = x_1$

The algorithm stops after two rounds.

$Q8$) We have n data points $x_1,..., x_n \in \mathbb{R}$. $y_1,..., y_n \in \mathbb{R}$ are the corresponding labels. We form a new dataset of n points given as :

$$D = \{(x_1, y_1),..., (x_n, y_n)\}$$

On running PCA on D, we get l_1 as the line along the first principal component.
On running Linear regression on D, we get l_2 which is of the form $y = wx$.

Assume $\sum_{i=1}^{n} (x_i, y_i) = (0, 0)$. Choose the correct options.

Abbreviations

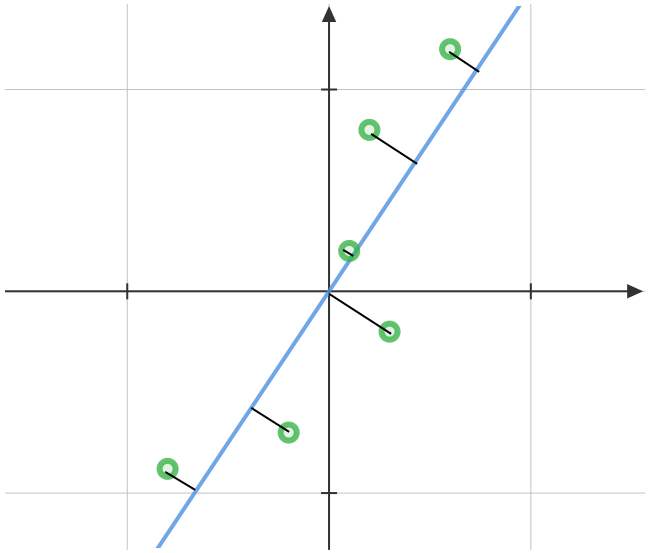
- SSPD - Sum Squared Perpendicular Distance
- SSVD - Sum Squared Vertical distance

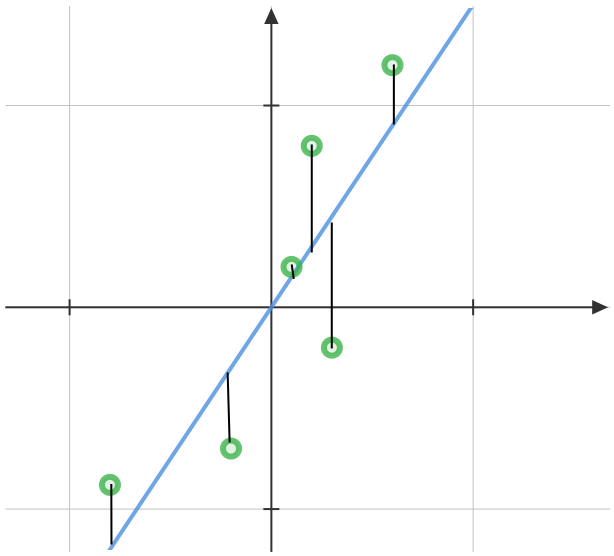
A. The SSPD from the data points in D to l_1 is the minimum possible SSPD to any line in $\mathbb{R}^2$
B. The SSPD from the data points in D to l_2 is the minimum possible SSPD to any line in $\mathbb{R}^2$
C. The SSVD from the data points in D to l_1 is the minimum possible SSVD to any line in $\mathbb{R}^2$
D. The SSVD from the data points in D to l_2 is the minimum possible SSVD to any line in $\mathbb{R}^2$

ANSWERS : A,D

PCA minimizes $||X - \left(X^t w\right) w||^2$

Linear Regression minimizes $||X^T w - Y||^2$





Left : Error minimization by PCA
Right : Error minimization by Linear Regression

QUESTION 9

Gaussian kernel regression with parameter $\sigma^2 = \frac{1}{2}$ was applied to the following dataset with two features:

$$X = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}, \quad y = \begin{bmatrix} 2.1 \\ 1 \\ 2 \\ 1.2 \end{bmatrix}$$

The weight vector can be written as $\mathbf{w} = \phi(X)\alpha$, where ϕ is the transformation mapping corresponding to the kernel. The vector α is given by $\begin{bmatrix} 2.1 \\ -2.1 \\ 3 \\ 0 \end{bmatrix}$, which is obtained as $(K)^{-1}y$, where K is the kernel matrix. What will be the prediction for the point $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$?

Solution:

1. **Gaussian Kernel Function:**

   The Gaussian (or RBF) kernel function is defined as:

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right)$$

   Given $\sigma^2 = \frac{1}{2}$, we have $\sigma = \frac{1}{\sqrt{2}}$.

2. **Prediction Formula:**

   To predict the output for a new point $x_{\text{new}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, we use:

$$\hat{y} = \sum_{i=1}^{4} \alpha_i K(x_{\text{new}}, x_i)$$

   where x_i are the points in the training set and $\alpha = \begin{bmatrix} 2.1 \\ -2.1 \\ 3 \\ 0 \end{bmatrix}$.

3. **Calculate Pairwise Distances:**

   We need to calculate the squared Euclidean distances $\|x_{\text{new}} - x_i\|^2$ between $x_{\text{new}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and each x_i in X:

   - $x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:

     $$\|x_{\text{new}} - x_1\|^2 = (1-1)^2 + (1-0)^2 = 0 + 1 = 1$$

   - $x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$:

     $$\|x_{\text{new}} - x_2\|^2 = (1-0)^2 + (1-1)^2 = 1 + 0 = 1$$

   - $x_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

     $$\|x_{\text{new}} - x_3\|^2 = (1-1)^2 + (1-1)^2 = 0 + 0 = 0$$

   - $x_4 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$:

     $$\|x_{\text{new}} - x_4\|^2 = (1-0)^2 + (1-0)^2 = 1 + 1 = 2$$

4. **Compute Kernel Values:**

   Substitute the squared distances into the Gaussian kernel formula with $\sigma^2 = \frac{1}{2}$:

$$K(x_{\text{new}}, x_1) = \exp\left(-\frac{1}{2 \times \frac{1}{2}}\right) = \exp(-1) = e^{-1}$$

$$K(x_{\text{new}}, x_2) = \exp\left(-\frac{1}{2 \times \frac{1}{2}}\right) = \exp(-1) = e^{-1}$$

$$K(x_{\text{new}}, x_3) = \exp\left(-\frac{0}{2 \times \frac{1}{2}}\right) = \exp(0) = 1$$

$$K(x_{\text{new}}, x_4) = \exp\left(-\frac{2}{2 \times \frac{1}{2}}\right) = \exp(-2) = e^{-2}$$

5. **Compute Prediction:**

   Using $\alpha = \begin{bmatrix} 2.1 \\ -2.1 \\ 3 \\ 0 \end{bmatrix}$, we get:

   $$\hat{y} = \alpha_1 K(x_{\text{new}}, x_1) + \alpha_2 K(x_{\text{new}}, x_2) + \alpha_3 K(x_{\text{new}}, x_3) + \alpha_4 K(x_{\text{new}}, x_4)$$

   Substitute the values:

   $$\hat{y} = 2.1 \cdot e^{-1} + (-2.1) \cdot e^{-1} + 3 \cdot 1 + 0 \cdot e^{-2}$$

   Simplify the expression:

   $$\hat{y} = 2.1 \cdot e^{-1} - 2.1 \cdot e^{-1} + 3 + 0 = 3$$

Final Answer:

The prediction for the point $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is:

$$\hat{y} = 3$$

QUESTION 10

Consider fitting a linear regression model (as stated below) for the following data:

x	y
-1	1
0	-1
2	1

Fit $y_i = \beta_0$. Find β_0.

Solution:

Since the model is $y_i = \beta_0$, it implies that y is constant for all values of x. The value of β_0 can be found as the mean of the y values in the given data:

1. Calculate the sum of y values:

$$\sum y = 1 + (-1) + 1 = 1$$

2. Calculate the mean of y:

$$\beta_0 = \frac{\sum y}{n} = \frac{1}{3} = \frac{1}{3}$$

Therefore, $\beta_0 = \frac{1}{3}$.